



DETERMINANTS

SINGLE CHOICE QUESTION

1. Let m be a positive integer &

$$D_r = \begin{vmatrix} 2r-1 & {}^m C_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix} \quad (0 \leq r \leq m),$$

then the value of $\sum_{r=0}^m D_r$ is given by:

- (A) 0 (B) $m^2 - 1$
(C) 2^m (D) $2^m \sin^2(2^m)$

2. There are two values of making the value of

the determinant $\begin{vmatrix} 1 & -2 & 5 \\ 2 & x & -1 \\ 0 & 4 & 2x \end{vmatrix}$ equal to 86.

The sum of these two numbers, is

- (A) -4 (B) 5
(C) -3 (D) 9

3. Let $f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 1+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 1+4\sin 2x \end{vmatrix}$, then

the maximum value of $f(x)$ =

- (A) 2 (B) 4
(C) 6 (D) 8

4. If $px^4 + qx^3 + rx^2 + sx + t = \begin{vmatrix} x^2 + 3x & x-1 & x+3 \\ x+1 & 2-x & x-3 \\ x-3 & x+4 & 3x \end{vmatrix}$

then t =

- (A) 33 (B) 0
(C) 21 (D) none

5. If α, β & γ are the roots of the equation $x^3 + px + q = 0$,

then the value of the determinant $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ =

- (A) p (B) q
(C) $p^2 - 2q$ (D) none

6. If the system of equations

$$x + 2y + 2z = 1$$

$$x - y + 3z = 3$$

$$x + 11y - z = b$$

has solutions, then the value of b lies in the interval

- (A) $(-7, -4)$ (B) $(-4, 0)$
(C) $(0, 3)$ (D) $(3, 6)$

7. The system of equations

$$kx + (k+1)y + (k-1)z = 0$$

$$(k+1)x + ky + (k+2)z = 0$$

$$(k-1)x + (k+2)y + kz = 0$$

has a non-trivial solution for

- (A) exactly three real values of k .
(B) exactly two real values of k .
(C) exactly one real value of k .
(D) infinite number of values of k .

8. If A, B and C are such that $A + B + C = 0$, then the

value of $\begin{vmatrix} 1 & \cos C & \cos B \\ \cos C & 1 & \cos A \\ \cos B & \cos A & 1 \end{vmatrix}$ is

- (A) 1
(B) $1 - \cos^2 A - \cos^2 B - \cos^2 C - 2\cos A \cos B \cos C$
(C) 0
(D) -1

9. The determinant $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0$ if:

- (A) x, y, z are in A.P. (B) x, y, z are in G.P.
(C) x, y, z are in H.P. (D) xy, yz, zx are in A.P.

10. Area of triangle whose vertices $(a, a^2), (b, b^2), (c, c^2)$ is $\frac{1}{2}$, and area of another triangle whose vertices are $(p, p^2), (q, q^2)$ and (r, r^2) is 4, then the value of

$$\begin{vmatrix} (1+ap)^2 & (1+bp)^2 & (1+cp)^2 \\ (1+aq)^2 & (1+bq)^2 & (1+cq)^2 \\ (1+ar)^2 & (1+br)^2 & (1+cr)^2 \end{vmatrix}, \text{ is}$$

- (A) 2 (B) 4
(C) 8 (D) 16

11. If the system of equations

$$\begin{cases} x - \lambda y - z = 0 \\ \lambda x - y - z = 0 \\ x + y - z = 0 \end{cases}$$

has a unique solution, then the range of λ is $R - \{a, b\}$. Then the value of $(a^2 + b^2)$ is:

- (A) 1 (B) 2
(C) 4 (D) 9

12. The value of $\begin{vmatrix} -1 & 2 & 1 \\ 3+2\sqrt{2} & 2+2\sqrt{2} & 1 \\ 3-2\sqrt{2} & 2-2\sqrt{2} & 1 \end{vmatrix}$ is equal to

- (A) zero (B) $-16\sqrt{2}$
(C) $-8\sqrt{2}$ (D) None of these

COMPREHENSION TYPE QUESTIONS

Let α, β be the roots of the equation $ax^2 + bx + c = 0$.

Let $S_n = \alpha^n + \beta^n$ for $n \geq 1$ and

$$\Delta = \begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$$

13. If $\Delta < 0$, then the equation $ax^2 + bx + c = 0$ has
(A) positive real roots (B) negative real roots
(C) equal roots (D) imaginary roots
14. If a, b, c are rational and one of the roots of the equations is $1 + \sqrt{2}$, then the value of Δ is
(A) 8 (B) 12
(C) 30 (D) 32
15. If $\Delta > 0$, then
(A) $f(1) > 0$
(B) $f(1) < 0$
(C) $f(1) = 0$
(D) cannot say anything about $f(1)$

INTEGER TYPE QUESTIONS

16. If $a_1, a_2, a_3, 5, 4, a_6, a_7, a_8, a_9$ are in H. P. and
 $D = \begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$, then the value of $[D]$ is (where
[$\cdot$] represents the greatest integer function)

17. Let $D_1 = \begin{vmatrix} a & b & a+b \\ c & d & c+d \\ a & b & a-b \end{vmatrix}$ and $D_2 = \begin{vmatrix} a & c & a+c \\ b & d & b+d \\ a & c & a+b+c \end{vmatrix}$

then the value of $\left| \frac{D_1}{D_2} \right|$, where $b \neq 0$ and $ad \neq bc$, is

18. If $\Delta = \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$, then the value of
 $(\Delta_{\max})/2$ is.....

19. If $\begin{vmatrix} x^n & x^{n+2} & x^{n+4} \\ y^n & y^{n+2} & y^{n+4} \\ z^n & z^{n+2} & z^{n+4} \end{vmatrix} =$
 $\left(\frac{1}{y^2} - \frac{1}{x^2} \right) \left(\frac{1}{z^2} - \frac{1}{y^2} \right) \left(\frac{1}{x^2} - \frac{1}{z^2} \right)$

then $-n$ is

20. If the system of linear equations
 $(\cos\theta)x + (\sin\theta)y + \cos\theta = 0$
 $(\sin\theta)x + (\cos\theta)y + \sin\theta = 0$
 $(\cos\theta)x + (\sin\theta)y - \cos\theta = 0$
is consistent, then find the number of possible value
of θ is.....(where $\theta \in [0, 2\pi]$)

ASSERTION REASON TYPE QUESTIONS

21. Assertion (A): If the matrix $A = \begin{bmatrix} 1 & 3 & \lambda+2 \\ 2 & 4 & 8 \\ 3 & 5 & 10 \end{bmatrix}$ is

singular, then $\lambda = 4$.

Reason (R): If A is a singular matrix, then $|A| = 0$.

- (A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false and R is true

22. Let A be a 2×2 matrix.

Assertion (A): $\text{adj}(\text{adj} A) = A$

Reason (R): $|\text{adj} A| = |A|$

- (A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false and R is true

23. **Assertion (A):** If every element of a third order determinant of value D is multiplied by 5, then the value of the new determinant is $125D$.

Reason (R): If k is a scalar and A is an $n \times n$ matrix, then $|kA| = k^n|A|$

- (A) Both A and R are true and R is the correct explanation of A
 (B) Both A and R are true but R is not the correct explanation of A
 (C) A is true but R is false
 (D) A is false and R is true

MULTICORRECT TYPE QUESTIONS

24. The system of equation

$$x + 3y + 2z = 6$$

$$x + ay + 2z = 7$$

$$x + 3y + 2z = b$$

has

- (A) unique solution, if $a = 2$ and $b \neq 6$.
 (B) infinitely many solution, if $a = 4$ and $b = 6$.
 (C) no solution, if $a = 5$ and $b = 7$.
 (D) no solution, if $a = 3$ and $b = 5$.

25. Which of the following determinant(s) vanish(es)?

(A)
$$\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$$

(B)
$$\begin{vmatrix} 1 & ab & \frac{1}{a} + \frac{1}{b} \\ 1 & bc & \frac{1}{b} + \frac{1}{c} \\ 1 & ca & \frac{1}{c} + \frac{1}{a} \end{vmatrix}$$

(C)
$$\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$$

(D)
$$\begin{vmatrix} \log_x xyz & \log_x y & \log_x z \\ \log_y xyz & 1 & \log_y z \\ \log_z xyz & \log_z y & 1 \end{vmatrix}$$

ANSWER KEY

1. (A)
2. (A)
3. (C)
4. (C)
5. (D)
6. (A)
7. (C)

8. (C)
9. (B)
10. (D)
11. (B)
12. (B)
13. (D)
14. (D)

15. (D)
16. (2)
17. (2)
18. (5)
19. (4)
20. (2)
21. (A)

22. (B)
23. (A)
24. (B, C, D)
25. (A, B, C, D)



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